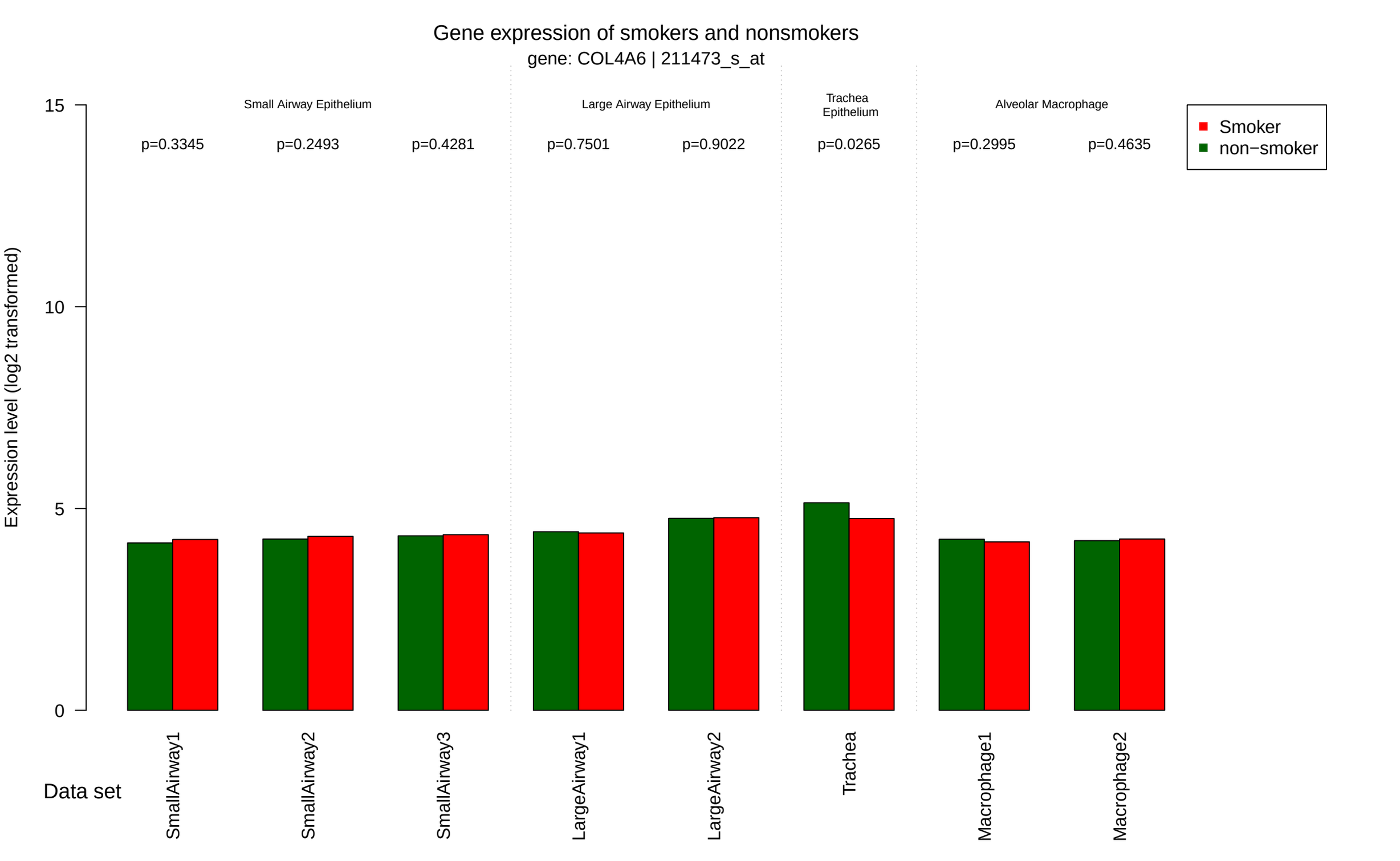
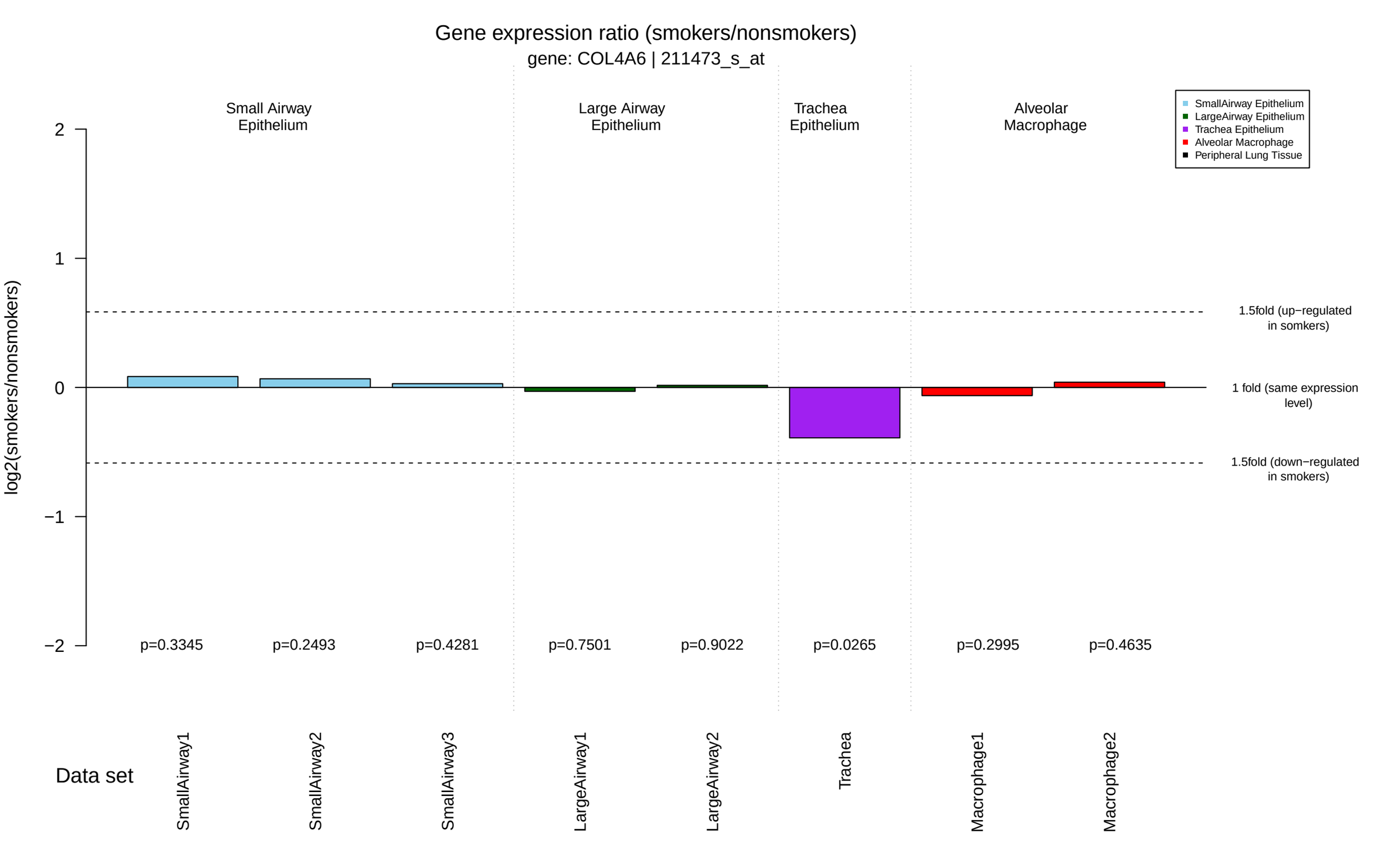
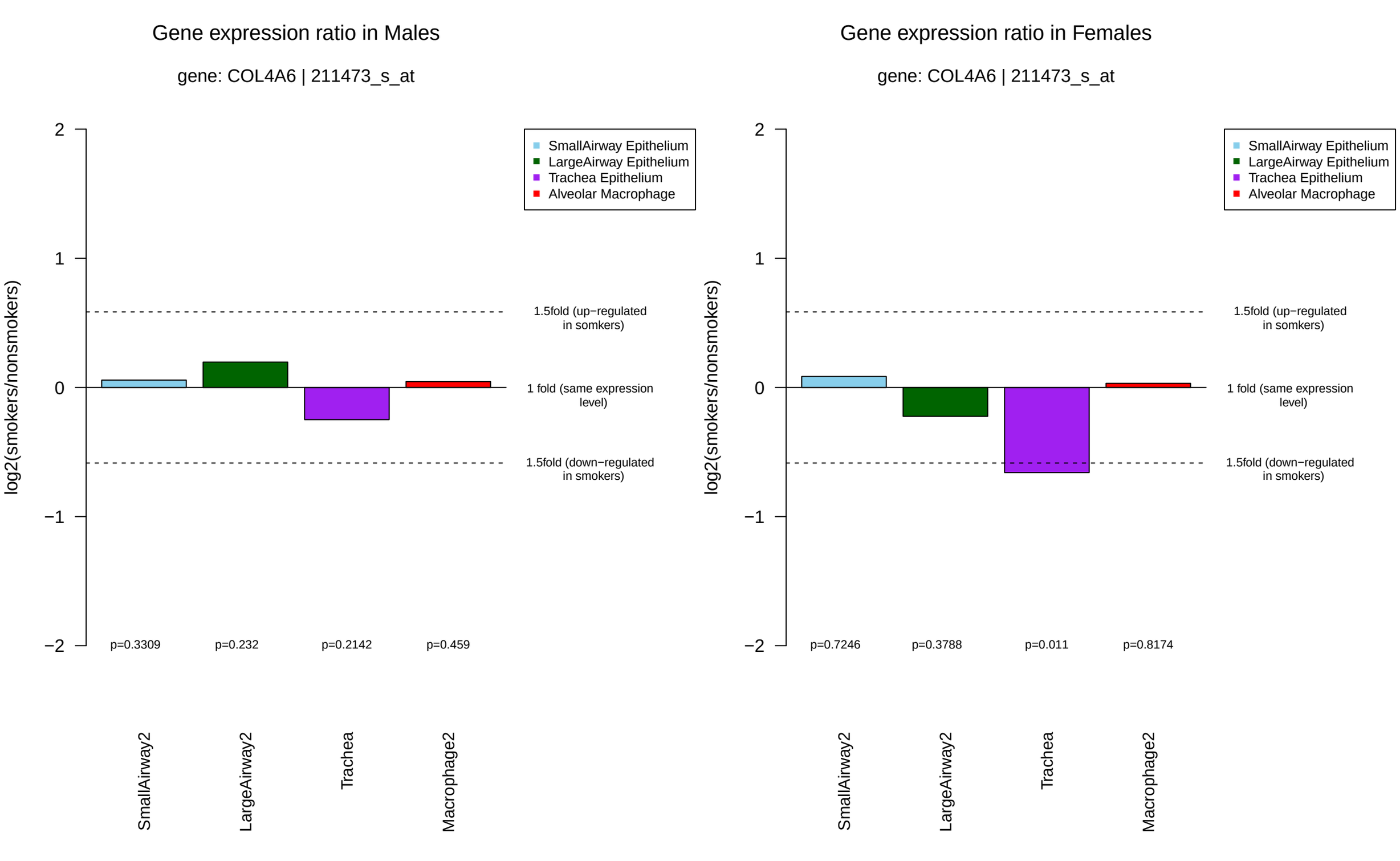
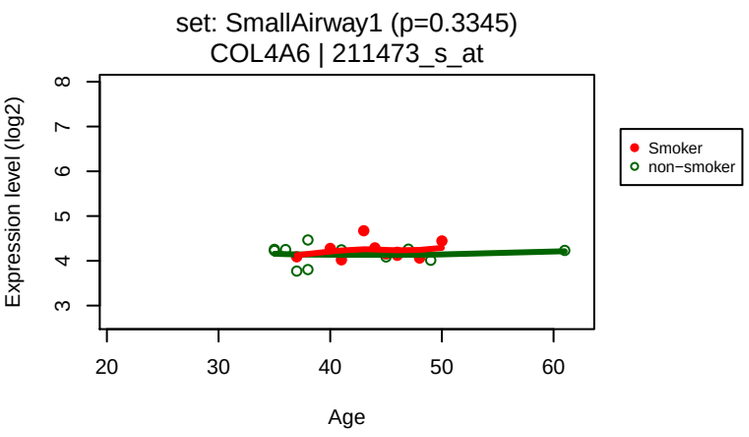
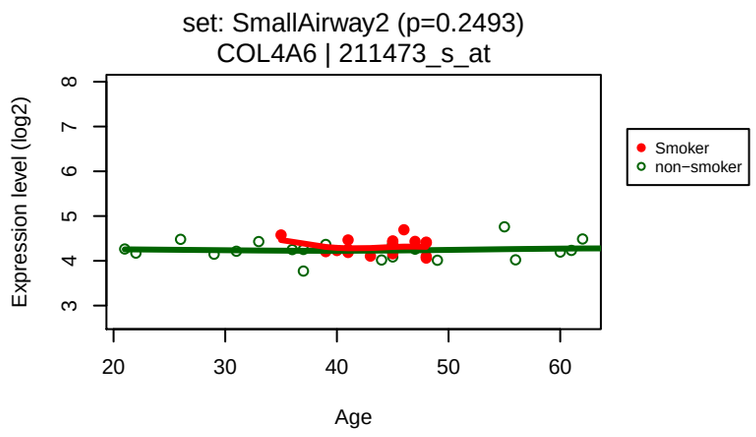
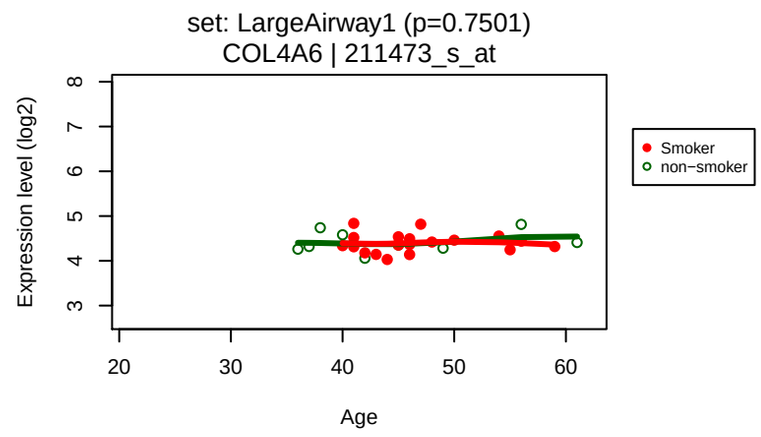

Gene expression of smokers and nonsmokers
gene: COL4A6 | 211473_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3345
p=0.2493
p=0.4281
p=0.7501
p=0.9022
p=0.0265
p=0.2995
p=0.4635
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

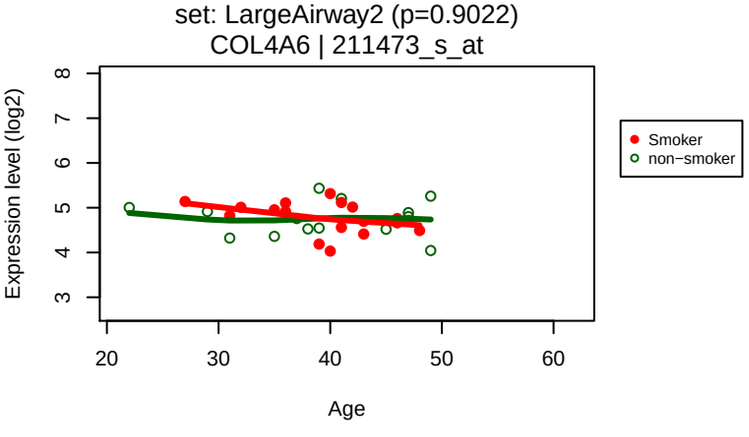

Gene expression ratio (smokers/nonsmokers)
gene: COL4A6 | 211473_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3345
p=0.2493
p=0.4281
p=0.7501
p=0.9022
p=0.0265
p=0.2995
p=0.4635
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

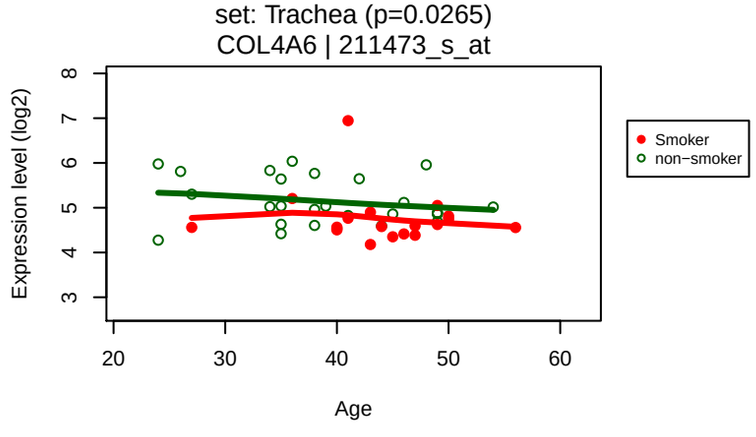

Gene expression ratio in Males
gene: COL4A6 | 211473_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3309
p=0.232
p=0.2142
p=0.459
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A6 | 211473_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7246
p=0.3788
p=0.011
p=0.8174
SmallAirway2
LargeAirway2
Trachea
Macrophage2

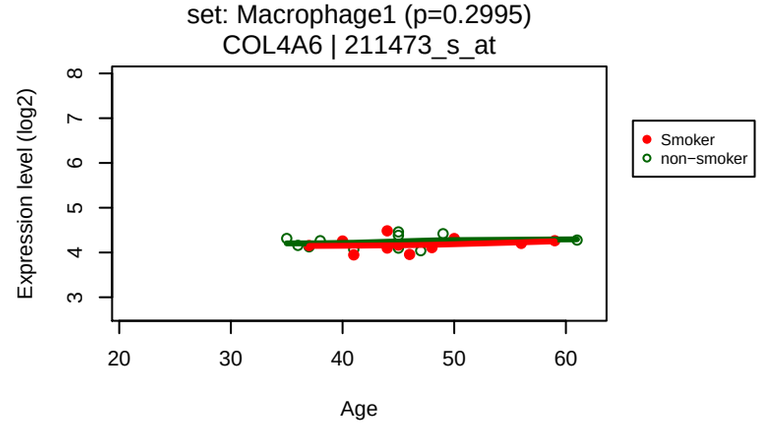

set: SmallAirway1 (p=0.3345)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker

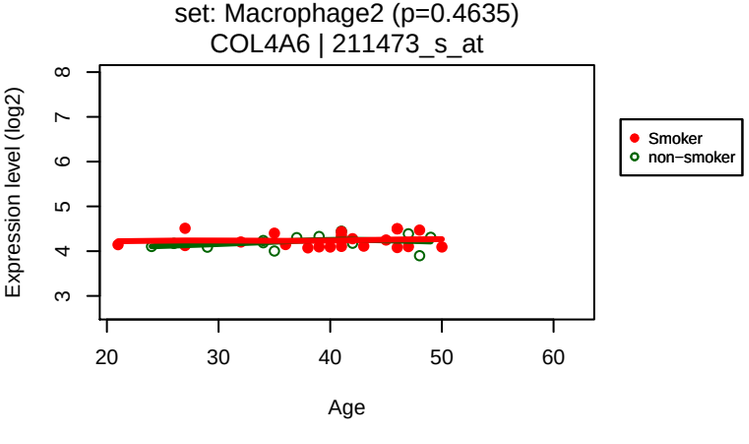

set: SmallAirway2 (p=0.2493)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker

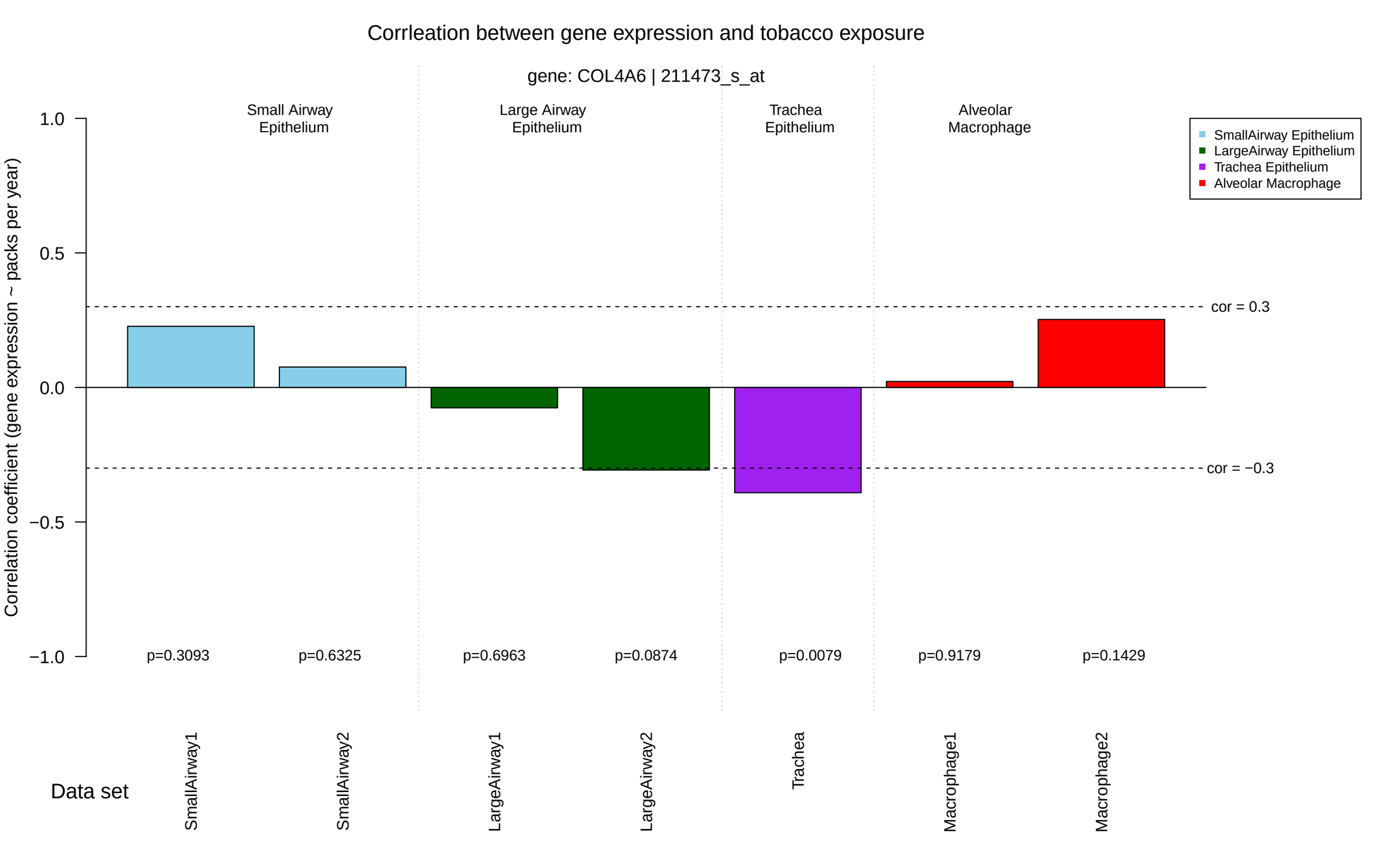

set: LargeAirway1 (p=0.7501)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9022)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0265)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2995)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4635)
COL4A6 | 211473_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A6 | 211473_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3093
p=0.6325
p=0.6963
p=0.0874
p=0.0079
p=0.9179
p=0.1429
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2